

Social and Cognitive Impairments in Rat Offspring after Ultrasound-Induced Prenatal Stress

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We studied the possibility of developing an autism model based on chronic prenatal psychological stress caused by variable frequency ultrasound 20-45 kHz. The offspring of female rats stressed during pregnancy demonstrated reduced time of social contacts in the social interaction test, increased anxiety in the open-field test, and memory impairment in the Morris water maze test in comparison with the control (intact) rat offspring. We also found a reducing trend in the *BDNF* gene expression in the amygdala in males of the experimental group. The results showed the possibility of developing the animal autism model based on prenatal stress.

Key Words: *prenatal stress; ultrasound effect; autism spectrum disorder; autism model; rat offspring*

Autism spectrum disorders (ASD) is a group of disorders that are a consequence of impaired development of the nervous system and characterized primarily by social dysfunction and stereotypic behavior [6]. The prevalence of this disease in the population is about 1% and is more common (by 4 times) in men than in women [6]. ASD is characterized by multifactorial etiology, heterogeneous symptomatology, as well as the absence of established general pathophysiology and effective treatment methods [3,11,13]. In this regard, the development of an experimental ASD animal model is an urgent task, because modeling of a neuropsychiatric disorder allows studying its mechanisms and possible treatment approaches [4]. Until now, there is no ASD animal model that reproduces all the features of this diseases group [11].

Numerous studies have demonstrated that prenatal stress plays an important role in the etiology of autism. In people, stressful events during pregnancy are associated with higher likelihood of developing autism

in a child [12]. The offspring of laboratory animals subjected to stress during pregnancy exhibited psychological and behavioral changes similar to ASD symptoms [14]. In this regard, we studied the possibility of developing an animal ASD model based on prenatal stress, which is based on pure mental trauma. Earlier in our Department of Basic and Applied Neurobiology, an experimental rat model of depressive disorder was developed, based on provide the chronic information uncertain conditions for animals by prolonged exposure to variable frequency ultrasound (US) [10,15].

The aim of this study was to assess the possibility of using offspring of rats subjected to US during pregnancy as an ASD model.

MATERIALS AND METHODS

We used Sprague Dawley rats from the Nursery for Laboratory Animals (Pushchino) in the experiment. The experiments were performed in accordance with Laboratory Practice Standards in the Russian Federation (Order No. 267 of Ministry of Health of the Russian Federation, April 1, 2016). Ten female rats were divided into two groups. The rats of the experimental group ($n=6$) were kept in individual cages after fertilization and throughout the pregnancy (21 days)

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were exposed to US (20–45 kHz) produced by an US generator placed at a distance of 1.5 m from the cage. Technical specifications of US generator: 220 V with adapter (0.3 W). After pregnancy, the dams with offspring were kept in individual cages under normal conditions without US. Control females ($n=4$) were kept under normal conditions in individual cages without US impact throughout pregnancy. After pregnancy, control dams with offspring were kept in individual cages (53×35×19 cm) under normal conditions.

At the age of 1 month, all rat pups were separated from dams (control and US-exposed), divided into males and females, and kept in common cages (≤ 10 rats per cage). Thus, two offspring groups were obtained for the experiment: experimental US group (19 males and 16 females) born by stressed mothers (US-offspring) and control group (8 males and 8 females) born by control mothers.

All pregnant female rats and offspring throughout the experiment were kept at a constant temperature (23°C) with controlled direct lighting (12/12 h) and free access to water and food.

At the age of two months, the rats were tested in behavioral tests in the following order: social interaction test, anhedonia test, Porsolt forced swimming test, open-field test, and Morris water maze test. All tests were conducted from 10.00 to 13.00. Animal behavior in all tests was recorded with a digital video camera and analyzed using RealTimer software (OpenScience). No more than one test per day was performed; the interval between tests was 2–3 days in all groups.

For the social interaction test, male and female rats were kept in individual cages for 2 days. At the beginning of the test, a juvenile (1-month-old) male or female rat was placed in the cage. Social contacts with a juvenile male and aggressive behavior of the test rat were recorded over 10 min. The percentage of rats demonstrating aggressive behavior was determined.

For the anhedonia test, male and female rats were placed in individual cages. On the first day, drinking bowls with a 1% sucrose solution were placed in each cage for familiarization. In 24 h, the drinking bowls with sucrose solution were removed and two drinking bowls were placed in each cage (1% sucrose solution and pure water). In 24 h, all the drinking bowls were pulled out. The volume of fluid consumed in both drinking bowls was measured. The sucrose preference index was calculated (% of consumed sucrose solution of the total volume of consumed fluid).

The open-field test was conducted on a square arena (45×45×41 cm). The bottom of the arena was divided into 9 squares (15×15 cm). The arena was illuminated with direct light of 100 lux. The arena was cleaned after each animal. The animal was placed in a

certain corner of the arena at the beginning of testing and its behavior was recorded over 5 min.

The Porsolt forced swimming test was used to evaluate despair-like behavior in rats. The rat was placed in a white plastic cylinder (height 40 cm, diameter 31 cm) with water so that the rat could not touch the bottom by the tail. The water temperature was 24°C. The rat has adapted for 2 min, the measurements were not made at this point. The immobility time was recorded during the next 6 min. Immobility was the absence of active movement. Water was changed after each rat.

The Morris water maze was used to evaluate the long-term spatial memory of the offspring. On the first day, the rats were trained to find a hidden under water platform in a round tank (diameter 1.8 m) filled with water. Each rat was placed eight times in the water for 1 min to platform search. Between the attempts, the rats were returned to the cage for 1 min. In 48 h after training, each rat was once placed in water and the time of finding the platform was measured. The test was limited to 1 minute. The water temperature was 24°C.

At the age of 3 months, the rats were deeply anesthetized with 5% ketamine solution (200 mg/kg intraperitoneally) and then decapitated; the brain was removed, the amygdala was isolated for subsequent evaluation of the expression of genes encoding brain-derived neurotrophic factor (BDNF) and oxytocin receptor (OXTR) by real-time PCR as described earlier [9]. β -Actin served as the reference gene. The relative expression of the target genes was calculated as follows: $2^{-\Delta\Delta Ct} \pm 2^{\pm SEM(\Delta Ct \text{ for each group})}$ [7].

The Statistica 6.0 software (StatSoft, Inc.) was used for statistical analysis. The normal distribution was verified by the Shapiro—Wilk test. In case of normal distribution, ANOVA was applied, and the results were presented as $M \pm SEM$; otherwise, Mann—Whitney U test was used, and the results were presented as $Me (Q1; Q3)$. The Spearman's rank correlation coefficient (r) was calculated for evaluation of the relationships between behavior and gene expression.

RESULTS

The social interaction test revealed a significant reduction in the time of social contacts of US-offspring in comparison with controls (Fig. 1). There was a significant difference in the total time of social contacts between US male rats and control male rats ($p=0.0008$), but we did not find significant differences in females ($p=0.14$). The total time of social contacts in the control males and females was 175.7 ± 24.7 and 78.7 ± 15.9 sec, respectively; in US-males and females, the corresponding values were $17.7 (8.1; 43.7)$ and 9.1

(3.2;79.6), respectively. Two US males demonstrated aggressive behavior (10.5% of the total number of rats in the US male group). Control animals did not demonstrate aggression to the juvenile male rat.

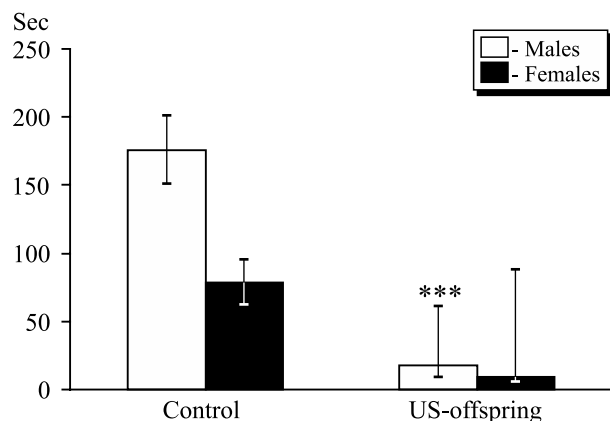


Fig. 1. The total time of social contacts in the offspring in the social interaction test. *** $p < 0.001$ in comparison with control male rats.

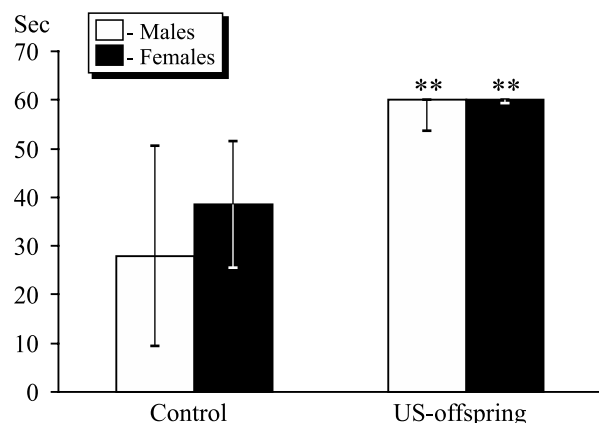


Fig. 2. Time of finding the platform in the Morris water maze. ** $p < 0.01$ in comparison with control males and females, respectively.

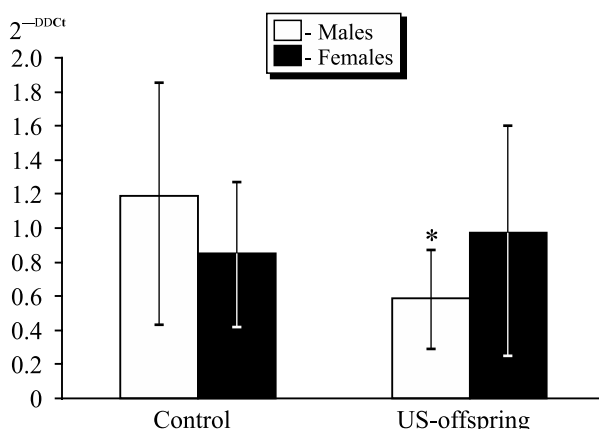


Fig. 3. *BDNF* gene expression in the amygdala of rat offspring at the age of 3 months. * $p = 0.049$ in comparison with control males.

The angedonia test and Porsolt forced swimming test demonstrated no significant differences between the control and experimental groups.

We did not find significant differences in the rearing frequency, grooming frequency, and total number of crossed squares in females and males in the open-field test, but freezing time in US females was significantly longer than in control females (28.1 (16.6; 15.2) and 0.3 (0.3; 1.7) sec, respectively; $p = 0.009$). In males, no significant differences in this parameter were found ($p = 0.07$). US females crossed the arena center less often than control females: 1 (0.25; 1) and 2.5 (1.5; 1.5), respectively ($p = 0.049$). At the same time, US males spent more time for grooming than control males: 30.1 (12.2; 10.6) and 11.9 (6; 4.8) sec, respectively ($p = 0.005$). The mean grooming duration in US males was also longer than in control males: 8.2 (3.8; 4) and 2.3 (1.8; 3.3) sec, respectively ($p = 0.02$).

The Morris water maze revealed a significant difference in the time of finding the platform between US-offspring and controls ($p = 0.006$ for males and $p = 0.008$ for females). These data indicate a significant spatial memory degradation in the US-offspring (Fig. 2). Most of the rats in the US-offspring group did not find the platform. At the same time, almost all control rats found the platform within one minute. The platform search time in the control offspring was 28 (18.5; 22.5) sec for males and 38.5 ± 13.0 sec for females; in the US-offspring group, the corresponding values were 60 (6.25; 0) sec for males and 60 (0.75; 0) sec for females.

When analyzing gene expression, we found that US males tended to have lower *BDNF* gene expression in the amygdala compared to control males ($p = 0.049$) (Fig. 3); in females, no differences in *BDNF* expression were found. Analysis of *OXTR* gene expression in the amygdala revealed no differences between the groups. A negative correlation was found between *BDNF* gene expression in the amygdala and total number of crossings the central square in the open-field in control males ($r = -0.82$, $p = 0.01$). Control females had a similar positive relationship ($r = 0.87$, $p = 0.005$). A positive correlation was found between *BDNF* gene expression level in the amygdala and the time of finding the platform in the Morris test for US males ($r = 0.79$, $p = 0.006$), as well as a negative correlation between *OXTR* gene expression and sucrose preference index in the angedonia test ($r = -0.64$, $p = 0.047$).

The results of our experiment demonstrated that the psychological prenatal stress offspring is characterized by a disturbance of social and cognitive functions and increase anxiety. At the same time, the US-offspring did not reveal anhedonia and despair-

like behavior, which are characteristic of animals in a depressive-like state. It indicates that the males and females from US-offspring do not have depression. At the same time, the degradation of social functions and memory and increase anxiety may indicate a disorder similar to ASD in the US-offspring. The clinical symptoms of ASD in humans are very heterogeneous. However, the main symptoms underlying the ASD diagnosis are deficit of social interactions and stereotypic behavior [3,6]. Possible concomitant symptoms include handicap mental, anxiety, attention deficit hyperactivity disorder [6]. As the diagnosis of ASD is based on behavioral phenotypes, animal ASD modeling requires the model to include one of the main ASD symptoms [3]. In our study, we demonstrated the presence of social deficiency in the experimental group, as well as increased anxiety in most animals. This already indicates compliance with the face validity criterion [1]. Thus, we provide for developing an autism model based on the psychological prenatal stress caused by US exposure.

We found a reducing trend in the amygdala *BDNF* gene expression from the adult US males. However, a similar decrease was demonstrated previously [2] and explained by changes in DNA methylation in exon 4 of the *BDNF* gene. It is known that the amygdala is involved in the processing of threat signals, including dangerous social signals [8], as well as in the process of emotional learning [13]. The expression of *BDNF* is enhanced and its receptors are activated in the amygdala as a response to fear [5]. It can be hypothesized that prenatal stress affects this reaction to a certain extent and disrupts behavior, including social behavior. However, minor changes in our study can be explained by the fact that *BDNF* expression was analyzed in adult animals. The amygdala and other brain areas have abnormal properties in autism. So, in ASD people, the amygdala is hypoactive in a number of social tasks [8]. It can be assumed that *BDNF* in the amygdala plays the part in such disorders. There are no published data on *BDNF* gene expression in the amygdala in autism in adults, however, *BDNF* expression in the amygdala can be impaired in ASD.

All of the above assumptions based on the data obtained require additional studies. In particular, expression of the above genes should be analyzed in newborns or juvenile animals. Additional behavioral tests are also needed to confirm the possibility of using prenatally stressed animals as the ASD model.

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